

ER Diagram Design Report

Project Title: E-Commerce Order Management Database System

Objective

Design an Entity Relationship (ER) Diagram and convert it into a Relational Schema for the E-Commerce Order Management Database System.

Entities

Customer, Category, Supplier, Product, Orders, OrderDetails, Payment, Shipment, Review.

Relationships

Customer-Orders (1:M), Category-Product (1:M), Supplier-Product (1:M), Orders-OrderDetails (1:M), Orders-Payment (1:1), Orders-Shipment (1:1), Customer-Review (1:M), Product-Review (1:M), Orders-Product (M:M through OrderDetails).

Participation Constraints

Each Order belongs to one Customer. Each Product belongs to one Category and one Supplier. Each OrderDetail belongs to one Order and one Product. Each Payment and Shipment belongs to one Order. Each Review belongs to one Customer and one Product.

Relational Schema Summary

Tables: CUSTOMER, CATEGORY, SUPPLIER, PRODUCT, ORDERS, ORDERDETAILS, PAYMENT, SHIPMENT, REVIEW with appropriate Primary Keys and Foreign Keys.

Conclusion

The ER Diagram and Relational Schema provide a structured logical database design that ensures data integrity, minimizes redundancy, and supports efficient implementation of the E-Commerce Order Management Database System.